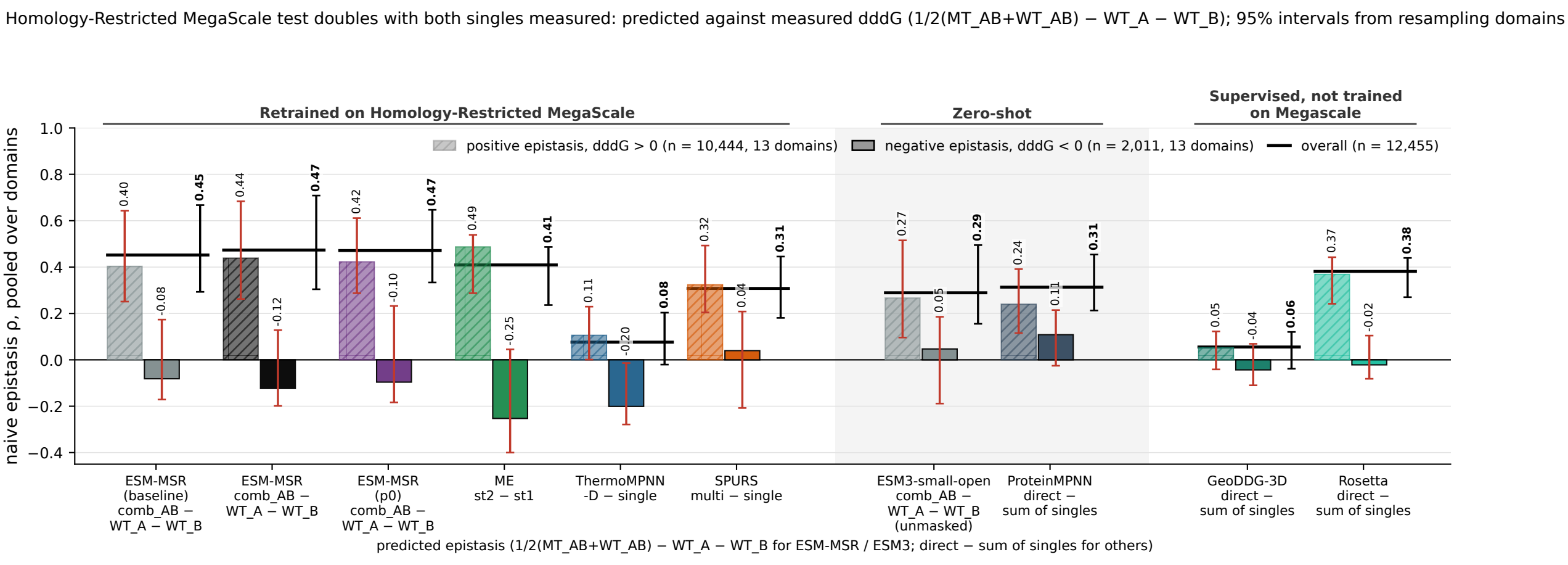
Homology-Restricted MegaScale test doubles with both singles measured: predicted against measured dddG (1/2(MT_AB+WT_AB) − WT_A − WT_B); 95% intervals from resampling domains
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
positive epistasis, dddG > 0 (n = 10,444, 13 domains)
negative epistasis, dddG < 0 (n = 2,011, 13 domains)
overall (n = 12,455)
naive epistasis ρ, pooled over domains
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
0.40
-0.08
0.45
0.44
-0.12
0.47
0.42
-0.10
0.47
0.49
-0.25
0.41
0.11
-0.20
0.08
0.32
0.04
0.31
0.27
0.05
0.29
0.24
0.11
0.31
0.05
-0.04
0.06
0.37
-0.02
0.38
ESM-MSR (baseline) comb_AB − WT_A − WT_B
ESM-MSR comb_AB − WT_A − WT_B
ESM-MSR (p0) comb_AB − WT_A − WT_B
ME st2 − st1
ThermoMPNN -D − single
SPURS multi − single
ESM3-small-open comb_AB − WT_A − WT_B (unmasked)
ProteinMPNN direct − sum of singles
GeoDDG-3D direct − sum of singles
Rosetta direct − sum of singles
predicted epistasis (1/2(MT_AB+WT_AB) − WT_A − WT_B for ESM-MSR / ESM3; direct − sum of singles for others)